# MD. MEHERAB HOSSEN

Mechanical Engineer | CFD & Thermal Sciences

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## Research Profile

Mechanical Engineering graduate (AUST, 2026) and CFD Simulation Engineer focused on multiphase flow, heat transfer, and phase change. Develop and validate numerical models in ANSYS Fluent and OpenFOAM against published benchmarks, with research in thermal energy storage and droplet dynamics.

## Education

**Ahsanullah University of Science and Technology (AUST)** **April 2022 – July 2026**

B.Sc. in Mechanical Engineering CGPA: 3.122/4.00

## Research Interests

Flow and heat transfer in porous media and subsurface systems; computational multiphase flow; phase-change heat transfer; benchmark-based numerical model validation. [Research evidence]

## Manuscripts & Conferences

**Md. Meherab Hossen**, Md. Asif Farhan, Tabasom Rahman, Hasibuzzaman Mahmud, Dewan Hasan Ahmed. "Numerical Investigation of Enhanced Thermal Energy Storage in Solid-Solid Phase Change Materials with New Fin Architecture for Shell-and-Tube Heat Exchanger Applications."
*Journal of Energy Storage*. First author; submitted October 2026; **under review**.

**10th BSME International Conference on Thermo-Fluid Engineering**
*Abstracts accepted; full papers submitted September 2026. Full-paper acceptance has not been confirmed.*

1. **Md. Meherab Hossen**, Md. Asif Farhan, Radia Siddika, Nowroze Farhan Ifraj. "Numerical Investigation of the Coupled Effect of Impact Angle and Ambient Rotor Downwash on Rain-Droplet Impact Regimes over UAV Surface." *First author.*
2. Md. Asif Farhan, **Md. Meherab Hossen**, Radia Siddika, Nowroze Farhan Ifraj. "Effect of Pre-Impact Thermal History on the Impact and Freezing of Free-Falling Supercooled Droplets on UAV-Relevant Surfaces." *Second author.*
3. Tabasom Rahman, Md. Asif Farhan, **Md. Meherab Hossen**, Fahmid Zaman Adnan, Ibrarur Rahman Choudhury, Dewan Hasan Ahmed. "Thermal Energy Storage Enhancement Using Alphabet-Inspired Fins in a Shell-and-Tube Heat Exchanger with Solid–Liquid PCM." *Third author.*

## Research Experience

**Undergraduate Research Assistant – Thermal Energy Storage** **2025 – 2026**

*Advisor: Prof. Dr. Dewan Hasan Ahmed, AUST*

- Developed Fluent phase-change models using temperature-dependent apparent heat capacity for the solid–solid transition at 25 °C; designed bio-inspired straight-fin arrangements.
- Validated liquid-fraction histories against Alnakeeb et al. and temperature histories against Farulla et al.; contributed to mesh checks using fully melted time in the solid–liquid PCM study.
- Analyzed five fin angles and orientations against an unfinned baseline in the solid–solid PCM study; the best simulated configuration reduced charging time by 79.3%.

## Research Experience (continued)

**Research Assistant – Droplet Dynamics** **2026 – Present**

*Research mentor: Nowroze Farhan Ifraj, PhD student in Mechanical Engineering, Virginia Tech*

- Configure VOF droplet models and analyze Weber-number effects on spreading and deformation.
- Reproduce Rafi et al. (2022) falling-droplet deformation and collision cases in ANSYS Fluent, with a spread-ratio error of 3%. [Validation]
- Benchmark OpenFOAM droplet-pair impact simulations against Goswami and Hardalupas (2023) over We = 54–128, with an 8% deviation from published data. [Validation] [Videos]

## Professional Experience

**CFD Simulation Engineer** **July 2026 – Present**

*MACPROTEC, Houston, Texas, USA | Remote*

- Analyze industrial dust collectors, baghouses and process pipelines using ANSYS Fluent, ANSYS Rocky and OpenFOAM.
- Develop 3D geometry in ANSYS DesignModeler and SolidWorks for simulation workflows.
- Reconstruct CAD geometry from laser scans using FARO SCENE, CloudCompare, MeshLab and Geomagic Design X.
- Configure discrete phase model (DPM) simulations of air–coal and air–dust particle transport.

## Thesis

**Experimental Characterization of *Nymphaea rubra* (Water Lily) Natural Fiber and Its Epoxy Composite through Mechanical Analysis** **July 2026**

Supervisor: Dr. Fazle Rabbi, AUST | Three-member research team

- Investigated water-lily fiber as a sustainable reinforcement for epoxy composites; conducted ASTM-based single-fiber tensile testing using a universal testing machine.
- Identified 48-hour retting as the best extraction condition. Composite properties: tensile strength $25.5 \pm 2$ MPa; elastic modulus $3.2 \pm 0.29$ GPa; flexural strength $22 \pm 5.5$ MPa. [Research] [Lab gallery]

## Projects

- Analyzed the **NACA 23012 airfoil** at Re $= 2 \times 10^6$ over angles of attack from $-5°$ to $24°$; compared results with Miley's (1982) wind-tunnel data. *Tools: XFLR5, Microsoft Excel.*
- Estimated 76,733 kWh/year and a 3.26-year payback for a 50 kW grid-connected PV plant. [Report]

## Skills

**CFD & simulation:** ANSYS Fluent, OpenFOAM, ANSYS Rocky, COMSOL.
**CAD & reverse engineering:** SolidWorks, ANSYS DesignModeler, FARO SCENE, CloudCompare, Geomagic Design X, MeshLab.
**Programming:** C, C++, MATLAB, Python.
**Scientific communication:** LaTeX, Microsoft Word, Excel (advanced), PowerPoint.

## Awards & Training

- Completed BITAC training in CNC/lathe machining, heat treatment, molding, and casting. [Certificate]
- Earned the **People's Choice Award – Mechanical Project Showcasing, Fall 2023.** [Portfolio]
- Completed **MATLAB Onramp – MathWorks.** [Certificate]

## References

**Prof. Dr. Dewan Hasan Ahmed**
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**Dr. Fazle Rabbi**
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